INCH POUND

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8 September 2025
SUPERSEDING
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DETAIL SPECIFICATION

CONVEYOR, ROLLER, GRAVITY, 1000 LB/FT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 SCOPE. This specification covers one type of roller conveyor.

2. APPLI CABLE DOCUMENTS

2. 1 General. The documents listed in this section are specified in section 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50271 - Plate, Identification

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-22248 - Weldments, Aluminum and Aluminum Alloy

(Copies of these documents are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue Philadelphia, PA 19111-5096 or e-mail to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2.2 Other Government Documents, Drawings, and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY COMBAT CAPABILITIES DEVELOPMENTCOMMAND SOLDIER CENTER DRAWINGS

11-1-589 - Conveyor, Roller Gravity, Airdrop, Aluminum, 1000 Lb/Ft Capacity - Assembly
11-1-590 - Frame - Assembly and Details
11-1-591 - Roller - Assembly and Details
11-1-592 - Coupling - Assembly and Details
11-1-593 - Shafts, Roller

(Copies of these documents are available from U.S. Army Combat Capabilities Development Command Soldier Center 10 General Greene Ave. Natick, MA 01760 or https://devcom.army.mil.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B20.1 - Safety Standard for Conveyors and Related Equipment

(Copies of this document are available from https://asme.org or ASME International, Three Park Avenue, New York, NY 10016-5990.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from https://asq.org or the American Society for Quality, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D5363 - Standard Specification for Anaerobic Single-Component Adhesives (AN)

(Copies of this document are available from www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, W. Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.2, 6.2, and 6.3).

3.2 Materials. All materials shall conform to the documents subsidiary to the drawings listed in section 2 and to the requirements specified herein.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.3 Construction. Construction and assembly shall be as specified herein and on the drawings listed in section 2. The conveyor shall conform to the applicable requirements of ASME B20.1.

3.3.1 Weldment. The aluminum frame weldment, as specified in Drawing 11-1-590, shall be fabricated by the inert-gas, tungsten-arc welding process and shall conform to the requirements applicable to class 3 weldments as specified by MIL-W-22248, except that the requirement for identification marking shall not apply. In lieu thereof, the part number and weldment serial number shall be applied in accordance with Drawing 11-1-590. Inspection marking requirements shall remain as specified.

3.3.2 Roller assemblies. The shell ends shall accommodate the bearings with the interference fit specified on the detail drawing. The bearings shall be free of lubricant. Prior to the insertion of the bearing into the shell, the entire surface area of contact shall be coated with sealing compound conforming to Group 01, Class 1, Grade 1 of ASTM-D5363. Exposed excess compound shall be removed after insertion.

3.3.2.1 Roller location. Unless otherwise specified, rollers shall be installed at the hexagonal hole locations at each end of the frame and at alternate hole locations thereafter.

3.4 Performance. The conveyor shall show no evidence of failure when tested as specified in 4.4.

3.5 Identification marking. An identification plate conforming to Composition A, Class 2 of A-A-50271 shall be the size and place in the location shown on Drawing 11-1-589.

3.6 Workmanship. Completed conveyor shall be free from dents, scratches, tool marks, splinters, sharp edges, welding voids, cracks and all evidence of rough handling or damage.

4. VERIFICATION

4.1 Classification of inspections. The inspections requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article is required (see 6.2), the gravity roller conveyor shall be examined as specified in 4.3.2 and 4.3.3. Any nonconformance shall be cause for rejection of the first article.

4.3 Conformance inspection. Unless otherwise specified, sampling for inspection shall be in accordance with ASQ Z1.4.

4.3.1 Component and material inspection. Materials and components shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase documents.

4.3.2 Intermediate inspection. Examination shall be made of aluminum frame weldments for visual defects, proper dimensions, and fluorescent penetrant. The sample unit shall be one aluminum frame weldment, as specified in Drawing 11-1-590. The inspection level shall be S-3. Any defect found shall be cause for rejection of the lot represented by the sample unit.

4.3.2.1 Visual and dimensional examination. Characteristics to be examined, defects to be scored, and acceptability standard shall be as prescribed for class 3 weldments by MIL-W-22248.

4.3.2.2 Fluorescent penetrant examination. The lateral support welds of each sample, selected at random, shall be examined in accordance with all requirements for methods, procedures and acceptability standards prescribed for class 3 weldments by MIL-W-22248.

4.3.3 End item inspection. Completed conveyors shall be examined for visual defects and proper dimensions. The sample unit shall be one complete conveyor, as specified in Drawing 11-1-589.

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4.3.3.1 Examination of visual characteristics. Characteristics to be examined and classification of defects shall be as listed in Table I. The inspection level shall be II.

TABLE I. Classification of visual defects.

Examine	Defect	Category	
		Major	Minor
Construction and workmanship	Any component missing, cracked, fractured, misaligned, or otherwise damaged, except as otherwise classified herein	101	
Hardware (screws, nuts, knobs and pins)	Missing, wrong type or size, stripped, or loose	102	
	Parts not finished as specified	103	
Roller assemblies	Shell dented, cracked, broken, or bearing not securely retained	104	
	Dirt, sealing compound, or other foreign matter in bearing preventing free rolling action	105	
Identification plate and marking	Plate, missing or not located as specified, incorrect or illegible		201
	Serial number or part number not as specified on Drawing 11-1-590		202

4.3.3.2 Examination of dimensions. Examination shall be made of dimensional characteristics pertaining to the assembled end item (i.e., overall frame dimensions, roller position intervals, location of lateral support and spacers and location of couplings). The inspection level shall be S-4. Each deviation from specified dimensions and tolerances shall constitute a defect.

4.4 Performance test. The sample unit for this test shall be one complete conveyor, Drawing 11-1-589. The conveyor shall be supported along the side rails at intervals of approximately 30 inches between centers. The bearing surface of each support shall provide approximately 4 square inches of surface contact with the side rails. A load of two thousand (2000) pounds shall be evenly distributed on a rigid bearing platform faced with 1/8 inch thick aluminum sheet. The bearing platform shall be constructed to dimensions approximately 20 inches wide by 30 inches long. The aluminum sheet facing shall span six rollers, center to center, providing the test load support of five rollers minimum, (approximately two (2) linear feet of conveyor). The load shall be cycled (rolled from one end of the conveyor to the other and return) five (5) times at a velocity of 175 (plus 10 or minus 5) feet per minute. The load shall be removed and the conveyor examined for defects indicated in Table II. The examination shall be conducted at inspection level S-2. Any defect shall be cause for rejection of the lot represented by the sample.

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TABLE II. Performance inspection.

Examine	Defect
Welds	Crack, break or other deformation
Roller assemblies	Dent, bend, bearing displaced from retaining dimples or not otherwise secure in shell, bound bearing, not otherwise free running roller, bent twisted or otherwise deformed shaft
Side rails, lateral supports, spacers and coupling	Twist, bend, crack, break or deformation

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from then managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The high capacity roller conveyor is intended for use in support of airborne operations and other load conveying applications involving loads not exceeding 1000 pounds per linear foot when the conveyor is supported at 30 inch intervals along the side rails. The conveyor is modular with provision for coupling.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of the specification.
b. When a first article is required (see 3.1, 4.2, and 6.3).
c. Number and location of rollers if other than specified (see 3.3.2.1).
d. Packaging requirements (see 5.1).

6.3 First article. When a first article is required, it should be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The first article should consist of the unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for inspection and approval of the first article.

6.4 Subject term (key word) listing.

Frame
Roller assemblies
Weldment

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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